

Week 1 — Model Selection (Review Gate #1)

- Survey 1–2 candidate LLM APIs (e.g., provider, model name, API endpoint style)
- Compare: price per 1K tokens (input/output), context length, latency (public claims), safety features, availability, SDK/docs, rate limits, region/compliance.
- Decide **primary** and **fallback** model; outline estimated monthly cost based on expected calls.
- Deliverables:
 - **1-page Purchase Proposal** (primary + fallback, rationale, risks).
 - **Comparison Table**

Week 2 — Scope, I/O & Generation Strategy (Review Gate #2)

- Define **input/output** JSON schema (de-identified only) and disclaimer line.
- Choose **generation** strategy: On-demand + cache (recommended) vs. batch precompute.
- Define **regeneration** triggers: fingerprint change (sorted JSON → SHA256), manual endpoint
- Deliverables: **1-page API spec** (schema + endpoints), **1-page strategy note** (cache & triggers), 5–10 sample inputs.

Week 3 — API Skeleton & First Model Call

- Build a simple endpoint `/v1/generate_story`.
- Implement JSON input validation and PHI (Protected Health Information) filters.
- Make a test call to the chosen LLM API using mock data.
Deliverable: endpoint runs locally and returns a story from sample input.

Week 4 — Prompt Refinement & Guardrails

- Improve story style, tone, and length through prompt engineering.
- Add basic output checks (length limit, PHI filter, banned words).
- Log model usage and latency.
Deliverable: stable story generation with guardrails; quality rubric draft. 1 page description of your sample story and explanation for clinicians review.

Week 5 — Evaluation, Docs & Final Demo

- Evaluate ≥30 cases on readability, factual alignment, safety; note failure modes.
- Optional endpoints: `GET /v1/patients/{id}/story`, `POST /v1/patients/{id}/regenerate`.
- README (setup, env/API keys, model choice rationale, generation strategy), short 3–5 min demo.

- Deliverables: **GitHub** submission (code + docs + sample stories) + **demo**.

All work will be rated upon the **Github** submission and **Google Drive Notes**